

EMRECAN GULAY

Research for complex AI and interactive systems

I study how people and AI systems shape an interaction, then use the findings to develop concepts, prototypes, and research directions.

PARTICIPANT STUDY

3 workshops · 22 qualitative researchers

WORKING ARTIFACT

Voice based desktop prototype

STATUS

Participant study complete · prototype evaluation pending

[live portfolio](#) | [email](#)

Relational Transparency Layer

Voice conversation with a local observer model

Conversation

local speech recognition and voice response

HUMAN

Analyze this fictional interview excerpt for tensions in discharge planning.

Nurse: The discharge plan says we should call within 48 hours, but in practice we are juggling missing medication lists, family members who heard different instructions, and patients who say they are fine before they feel safe enough to ask.

AI SYSTEM

I would code three tensions:

1. formal protocol colliding with fragmented handoff work;
2. relational safety becoming a condition for good assessment;
3. quick reassurance masking uncertainty after discharge.

HUMAN

That feels a bit too clean. Separate institutional pressure from what the nurse is doing in the interaction.

AI SYSTEM

The institutional pressure is the timed follow up. The interactional work is quieter: checking whether the patient can admit confusion without feeling blamed.

observer running

fictional healthcare interview

4 exchanges

Relational layer

updated after each exchange

Relational oscillations



Observer narrative

latest turns

Exchange 1

The participant sets the analytic frame and keeps the transcript visible. The AI receives a bounded task rather than an open conversation.

Exchange 2

The participant moves from asking for codes to checking the analysis. The AI holds a trainer stance and offers a tidy structure.

Exchange 3

The repair move changes the tone. The participant asks for a finer separation, and the AI shifts toward a narrower reading.

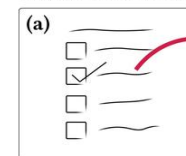
Exchange 4

The AI narrows its claim from broad coding to situated interpretation. The question of who carries the analytic judgement stays active.

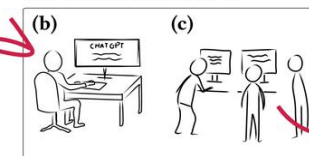
Episode 1 UNITS 1 TO 4

PARTICIPANT STUDY II WORKING PROTOTYPE

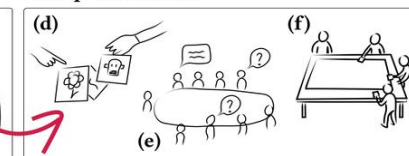
Pre-Session Survey



Individual Interaction

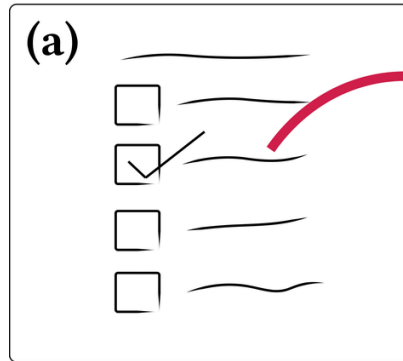


Group Interaction

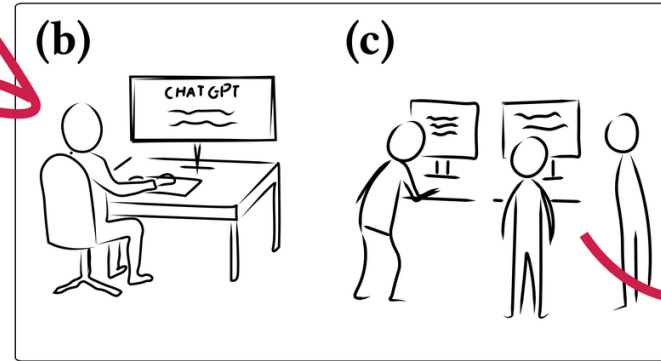


How do researchers describe AI, and how does that relation appear in their interaction?

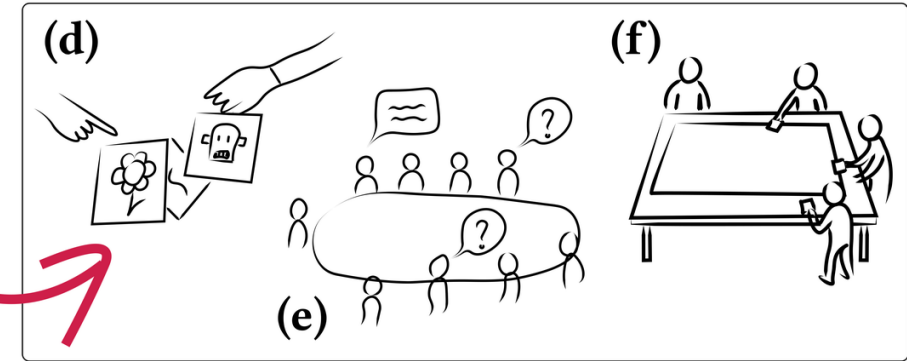
Pre-Session Survey



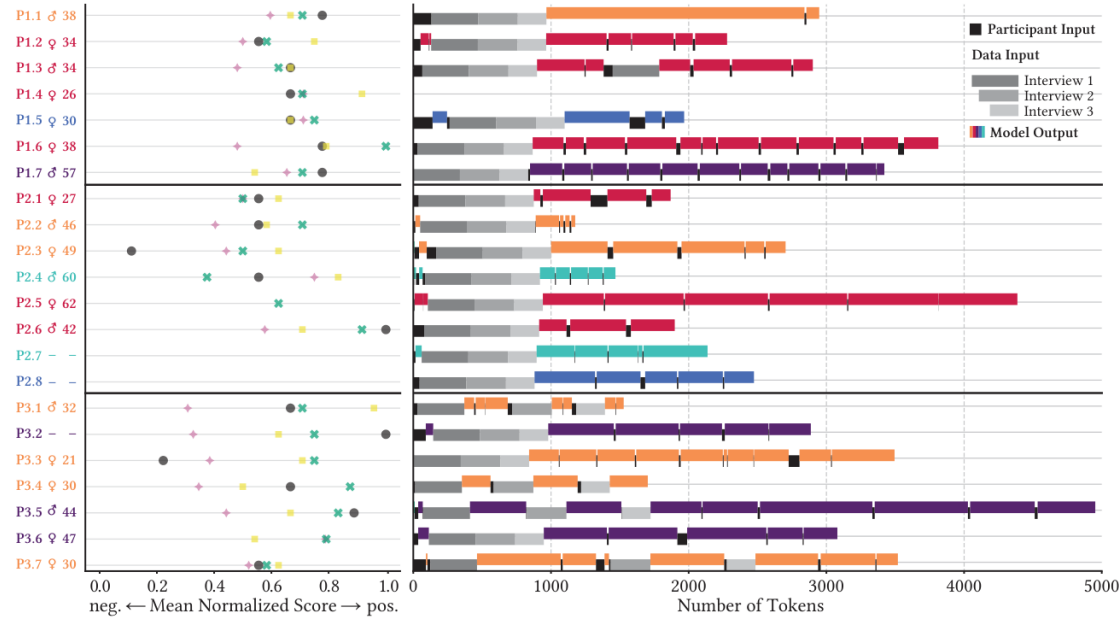
Individual Interaction



Group Interaction

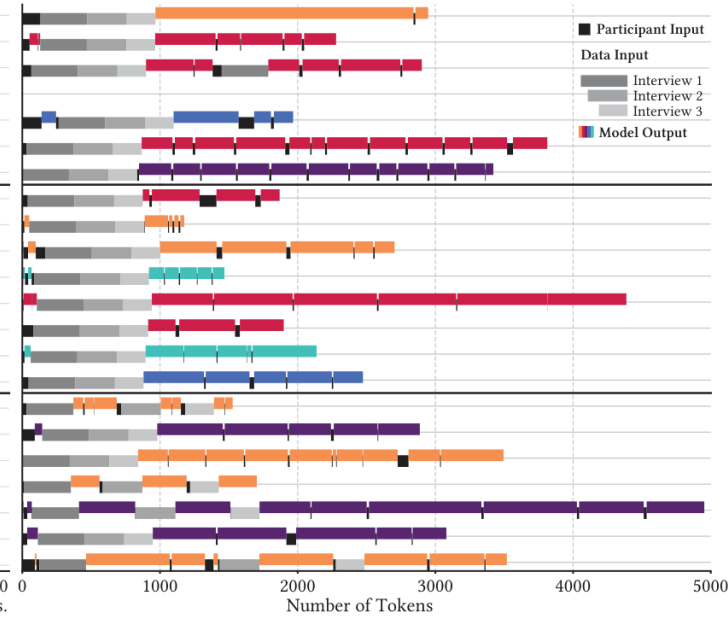


(a) Responses to Pre-Session Survey



Variable Group (n)
 Performance Expectancy (6)
 Effort Expectancy (6)
 Attitudes toward Use (13)
 Satisfaction (1)

(b) Chatlogs from Qualitative Research Task



EVIDENCE

DATA

Pre session surveys, chat logs, image choices, roundtable discussion, group worksheets, and observations.

ANALYSIS

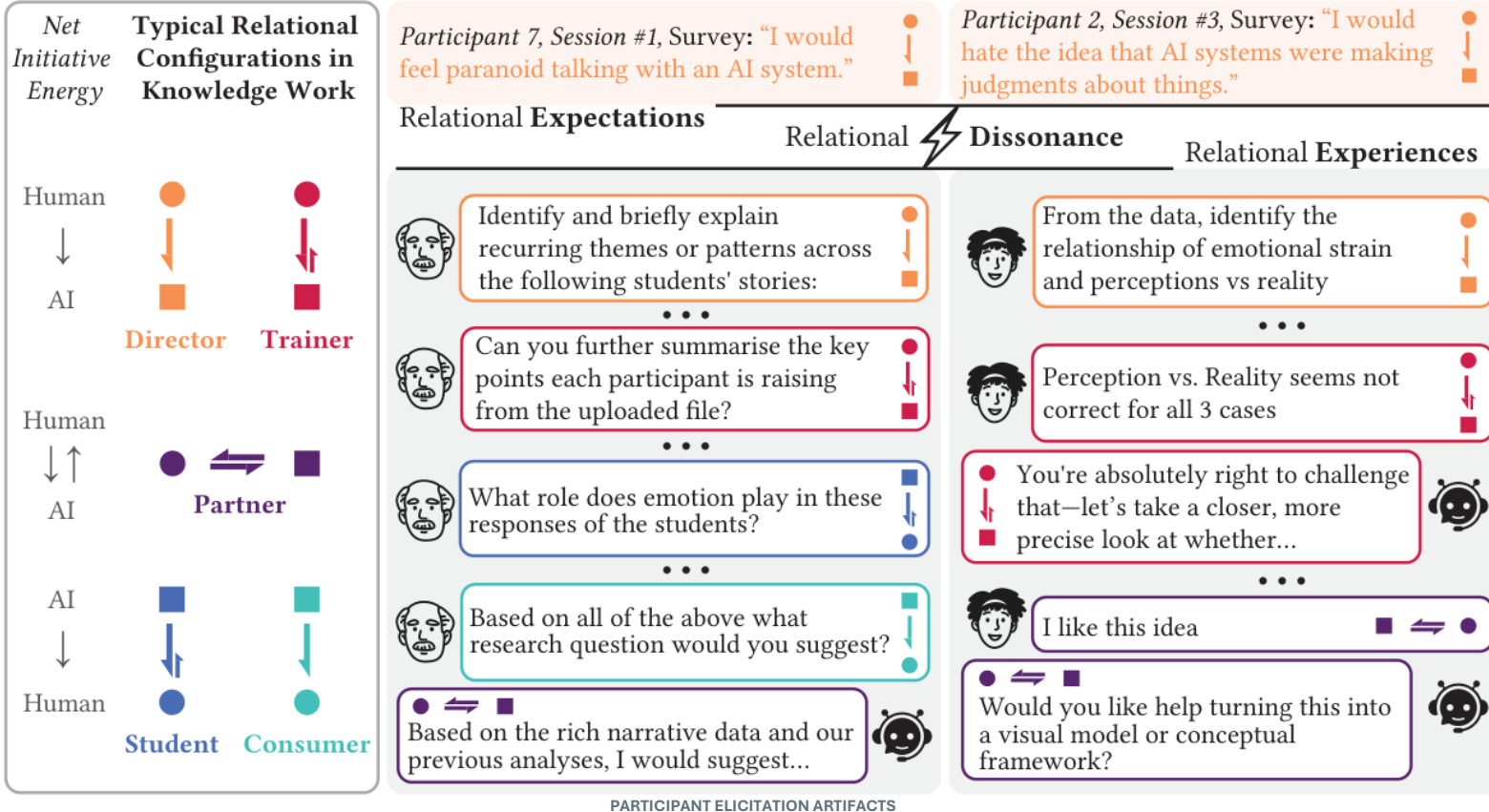
One analyst conducted the individual level interpretative phenomenological analysis; another conducted the social level thematic analysis. The team then integrated the findings.

ROLE

First author in a four person interdisciplinary research team.

3 workshops 22 participants 4 AI systems

Evidence basis: descriptive survey data, interaction traces, workshop material, and observations.



SYNTHESIS

Our analysis found that participants' stated position toward AI often differed from the relation enacted in their chat logs.

Five recurring configurations captured how role, control, and responsibility shifted across the work.

DESIGN RESPONSE

The findings informed Emreca's later Relational Transparency project and the prototype shown next.

Images in the lower strip were selected and discussed by participants during the workshop.



P1.1 P2.3 P3.7



P3.3 P1.2 P2.7



P2.2 P3.4



P1.3 P2.1



P3.3 P1.5



P3.4 P1.7



P3.2 P2.8

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AI SYSTEM

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Listening for the next spoken turn

Ready

Relational layer

updated after each exchange

Relational oscillations



Observer narrative

latest turns

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Episode 1 UNITS 1 TO 4

DIRECTOR to TRAINER mixed dynamic

V

03 WORKING PROTOTYPE

Relational Transparency Layer

A voice based desktop prototype that places the conversation, a provisional model account, and model generated relational indicators in one inspectable view.

ROLE

Principal investigator and sole demo paper author.

SEPARATE

Conversation and observation run as separate processes.

TRACE

The account is linked to completed turns and visible interaction evidence.

QUESTION

A demo user can compare the observer account with the transcript and their own experience.

BOUNDARY

Working demo. Formal participant evaluation remains future work.

WATCH THE 55 SECOND DEMO

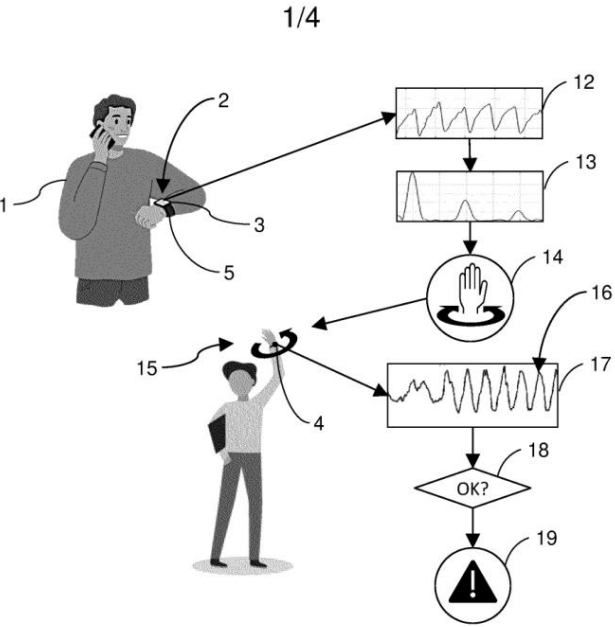


FIG. 1

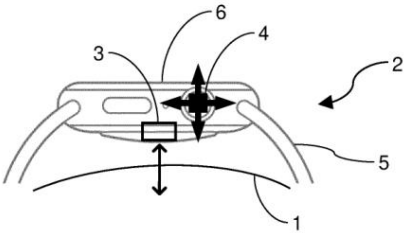


FIG. 2

| guided interaction for wearable sensing accuracy

WO2024235471A1

HUMAN MACHINE INTERACTION

HMI research and technology planning

HMI Technology Planning Specialist and HMI Research Intern at Huawei, 2022 to 2023.

PUBLIC RECORD

Named co inventor on five published international patent applications under the PCT.

- WO2024235471A1 wearable sensing accuracy
- WO2024213244A1 distributed smart devices
- WO2024188442A1 multisensory smart insole
- WO2024179676A1 communication haptics
- WO2024179677A1 activity related haptics

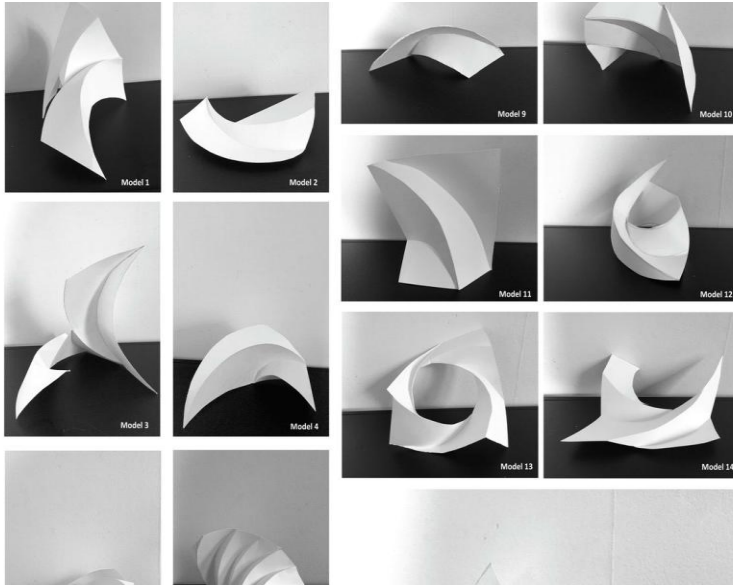
EVIDENCE BOUNDARY

Public records confirm co inventorship and publication of the five PCT applications. Specific research methods, decision authority, implementation, adoption, and product outcomes are unclaimed.

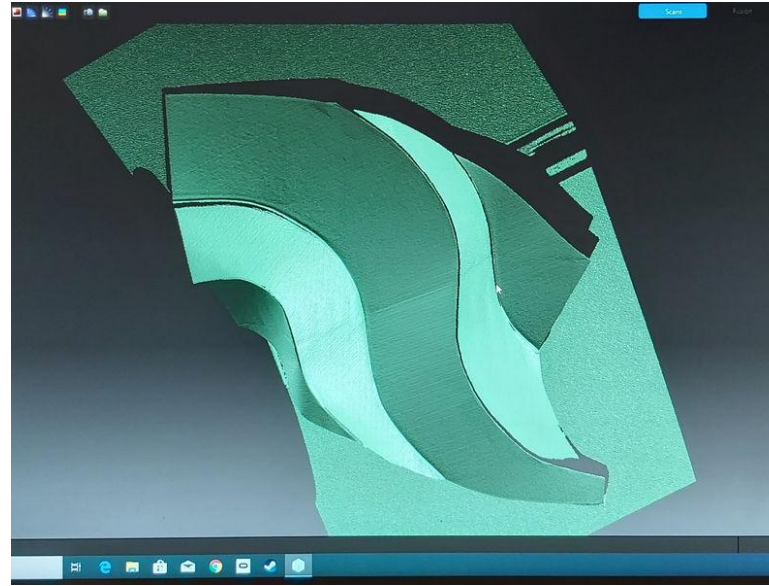
EMPLOYMENT RECORD SCALE (CV)

30+	5	11+	6
technology reports	PCT applications	research institutions	companies

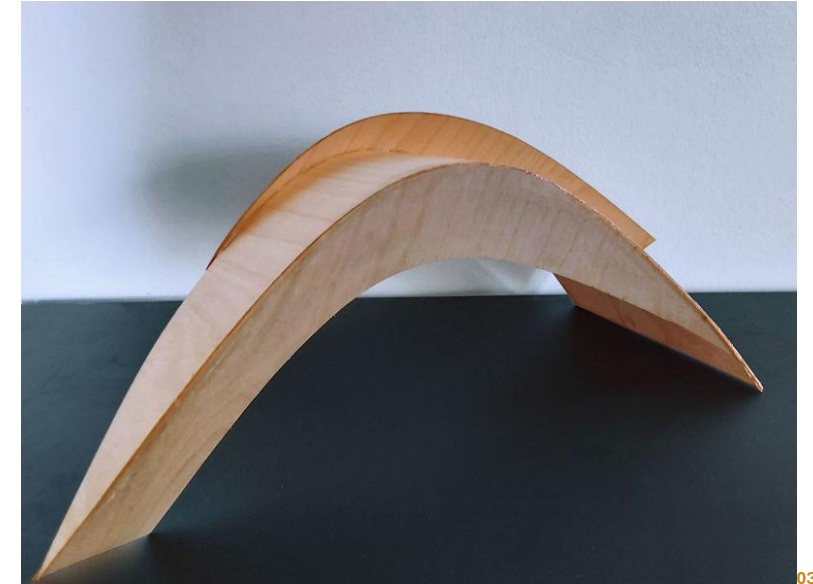
A design process moved between paper models, scanning, simulation, and plywood fabrication.



Explore and select



Translate and test



Fabricate and inspect

42

paper models

5

shortlisted

2

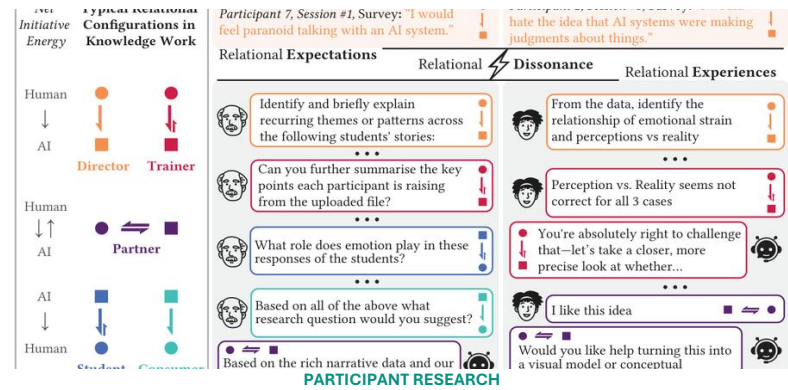
plywood prototypes

CHI 2021

verified publication

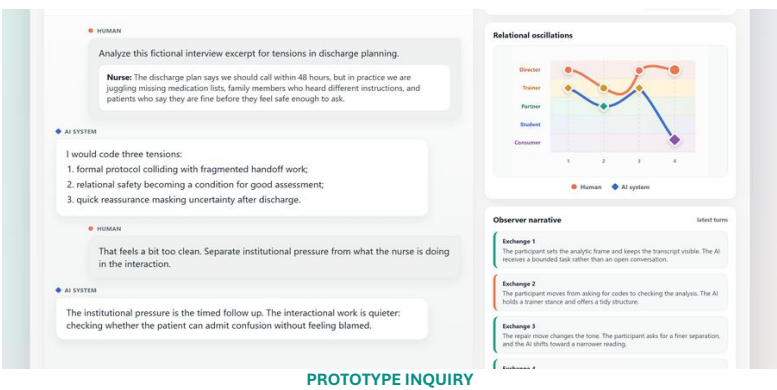
Role: first author; the paper states that the design process was mainly carried out by the first author.

Exploring a Feedback-Oriented Design Process Through Curved Folding | DOI [10.1145/3411764.3445639](https://doi.org/10.1145/3411764.3445639)



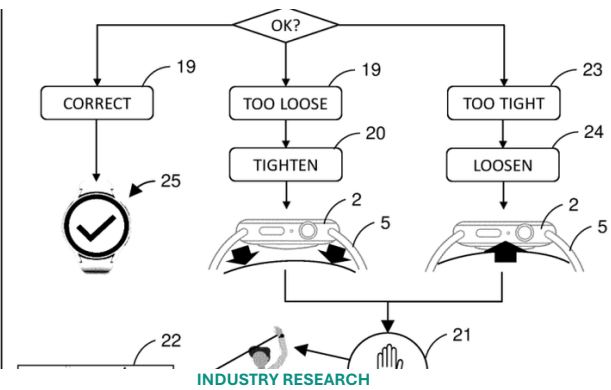
Study and synthesis

3 workshops · 22 researchers · multiple qualitative evidence sources



System behaviour

Working voice demo · provisional observer account · evaluation pending



Public invention record

Huawei · 5 published PCT applications · named co inventor

RESEARCH PRACTICE

I work across applied and academic settings where AI behaviour, handoffs, errors, repair, and responsibility need careful study.

Frame the phenomenon with participants · build an evidence base · develop artifacts that collaborators can question

EMRECAN GULAY

Doctor of Arts · Espoo, Finland

Concurrent Data Agent · Aalto School of Business · research data management and Open Research Network

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Live research portfolio

emrecangulay.com

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Google Scholar